Assignment: module -5 Network Fundamentals and Building Networks

Section 1: Multiple Choice

1. **What is the primary function of a router in a computer network?**
   Answer: c) Forwarding data packets between networks

2. **What is the purpose of DHCP (Dynamic Host Configuration Protocol) in a computer network?**
   Answer: d) Dynamically assigning IP addresses to devices

3. **Which network device operates at Layer 2 (Data Link Layer) of the OSI model and forwards data packets based on MAC addresses?**
   Answer: b) Switch

4. **Which network topology connects all devices in a linear fashion, with each device connected to a central cable or backbone?**
   Answer: b) Bus

Section 2: True or False

5. **A VLAN (Virtual Local Area Network) allows network administrators to logically segment a single physical network into multiple virtual networks, each with its own broadcast domain**.
   Answer: True

6. **TCP (Transmission Control Protocol) is a connectionless protocol that provides reliable, ordered, and error-checked delivery of data packets over a network.**
   Answer: False

7. **A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.**
   Answer: True

Section 3: Short

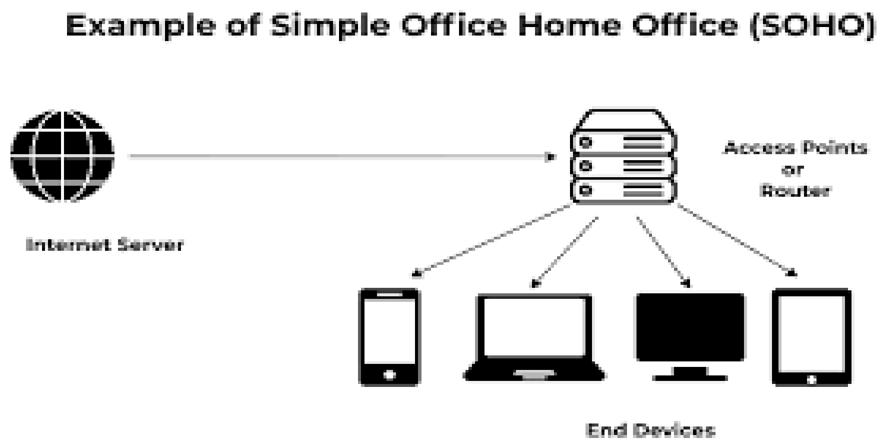
8. Describe the steps involved in setting up a wireless network for a small office or home office (SOHO) environment.

Setting up a wireless network in a SOHO environment involves several key steps:

1. **Choose the Right Equipment:**
   Select a wireless router that meets the size and performance needs of the office or home. Ensure it supports modern standards (like Wi-Fi 5 or Wi-Fi 6) for better speed and reliability.

2. **Connect the Hardware:**
   Connect the router to the internet source (modem or ISP line) using an Ethernet cable. Power on the router and ensure all indicator lights show normal operation.
3. **Access Router Configuration Page:**
   Use a computer or smartphone to connect to the router (via default Wi-Fi or cable). Open a web browser and enter the router's IP address (such as 192.168.1.1) to access the configuration settings.
4. **Configure Network Settings:**
   Set a unique network name (SSID) to identify your wireless network. Choose the appropriate region and wireless channel if required.
5. **Secure the Network:**
   Enable strong security such as WPA2 or WPA3 encryption. Set a strong password to prevent unauthorized access.
6. **Set Up DHCP and IP Settings:**
   Ensure DHCP is enabled so that devices automatically receive IP addresses. Configure IP range if necessary.
7. **Connect Devices:**
   Connect laptops, smartphones, printers, and other devices using the SSID and password.
8. **Test the Network:**
   Check internet connectivity on all devices. Test signal strength in different areas and reposition the router if needed.
9. **Optimize and Maintain:**
   Update router firmware regularly, change passwords periodically, and monitor connected devices to maintain security and performance.

These steps ensure a secure, efficient, and reliable wireless network for a SOHO environment.



Section 4: Practical

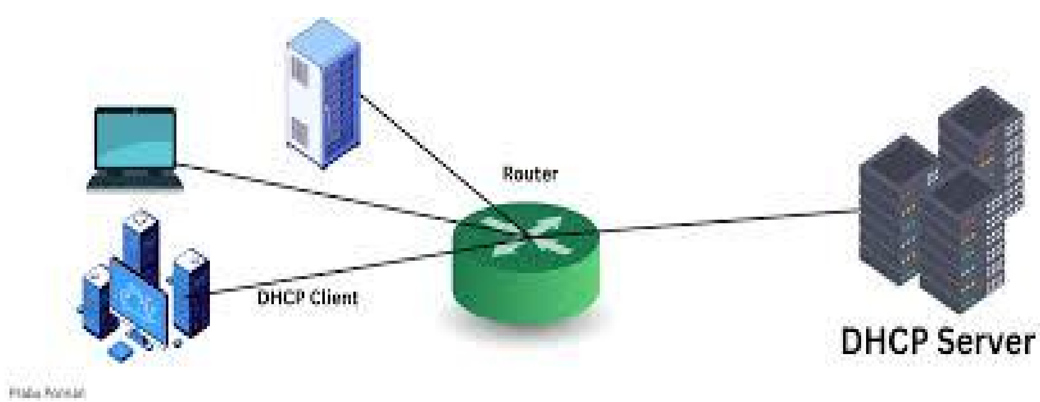
9. Demonstrate how to configure a router for Internet access using DHCP (Dynamic Host Configuration Protocol).

To configure a router for Internet access using DHCP, follow these steps:

1. **Connect the Hardware:**
   - Connect the router's WAN (Internet) port to the modem or ISP line using an Ethernet cable.
   - Connect a computer to the router using a LAN port or Wi-Fi.
2. **Access the Router Interface:**
   - Open a web browser and enter the router's default IP address (e.g., 192.168.1.1 or 192.168.0.1).
   - Log in using the default username and password (provided on the router label or manual).
3. **Configure WAN (Internet) Settings:**
   - Navigate to the "Internet" or "WAN" settings page.
   - Select **DHCP (Automatic IP)** as the connection type.
   - This allows the router to automatically obtain an IP address, subnet mask, gateway, and DNS from the ISP.

4. **Enable DHCP Server on LAN:**
   - Go to the LAN or DHCP settings section.
   - Enable the DHCP server so the router can assign IP addresses to connected devices.
   - Set an IP address range (e.g., 192.168.1.100 to 192.168.1.200).
5. **Save and Apply Settings:**
   - Click "Save" or "Apply" to store the configuration.
   - The router may reboot to apply changes.
6. **Test Internet Connection:**
   - Connect a device to the network.
   - Ensure it receives an IP address automatically.
   - Open a website to verify internet access.
7. **Verify Status:**
   - Check the router's status page to confirm it has received a valid IP address from the ISP.
   - Ensure DNS and gateway information are correctly assigned.

This process enables the router to automatically obtain network configuration from the ISP and provide seamless internet access to all connected devices.



Section 5: Essay

10. Discuss the importance of network documentation in the context of building and managing networks.

Network documentation is the process of recording all important information about a computer network, such as network design, devices, IP addresses, connections, and security settings.

It is very important in building and managing networks because it helps administrators understand how the network is organized. With proper documentation, it becomes easier to identify how devices like routers, switches, computers, and printers are connected.

One of the main benefits is **easy troubleshooting**. If a network problem occurs, administrators can quickly find the issue by checking the network diagram or records, which saves time and reduces downtime.

It also helps in **network management**. Administrators can easily monitor the network, make changes, and upgrade systems without confusion or errors.

Another important point is **security**. Network documentation keeps records of IP addresses, user access, and security settings, which helps in preventing unauthorized access and maintaining safety.

It is also useful for **future expansion**. When new devices need to be added, documentation helps in planning the network properly without affecting existing systems.

Finally, it supports **knowledge sharing**. If a network administrator leaves, new staff can easily understand the system using the documentation.

Conclusion

Network documentation is essential for making networks easier to manage, more secure, and more reliable. It plays a key role in building, maintaining, and expanding computer networks effectively.